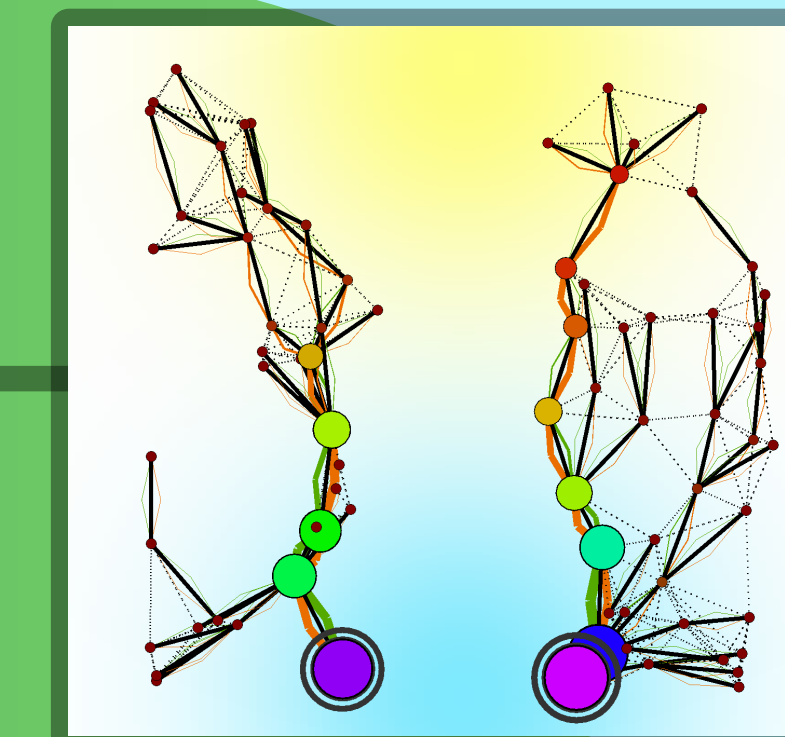
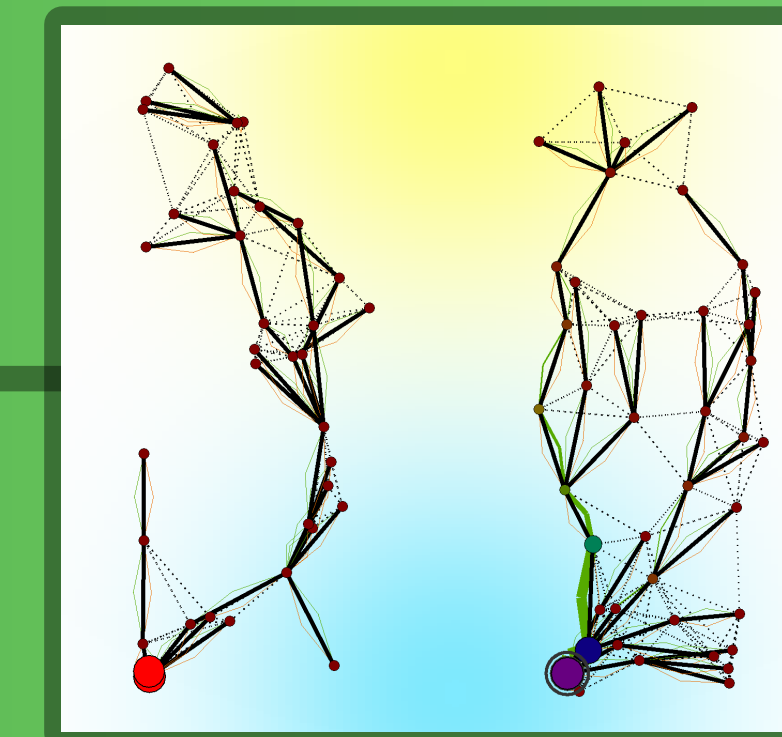
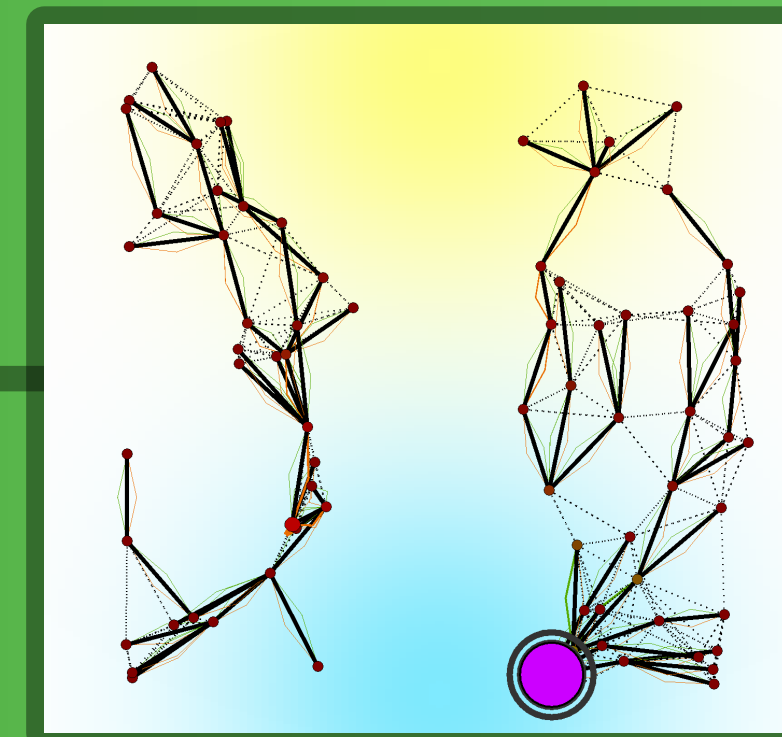
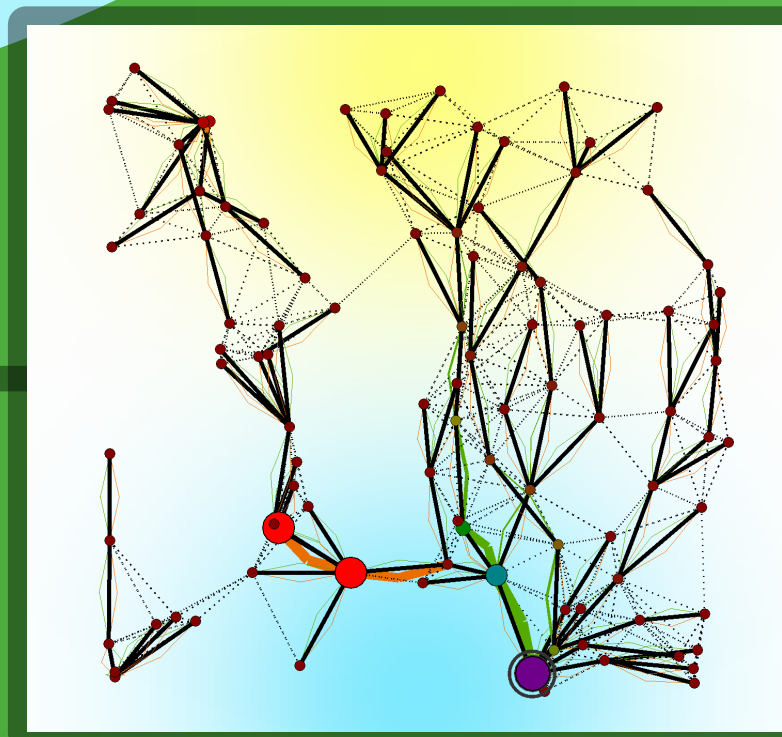
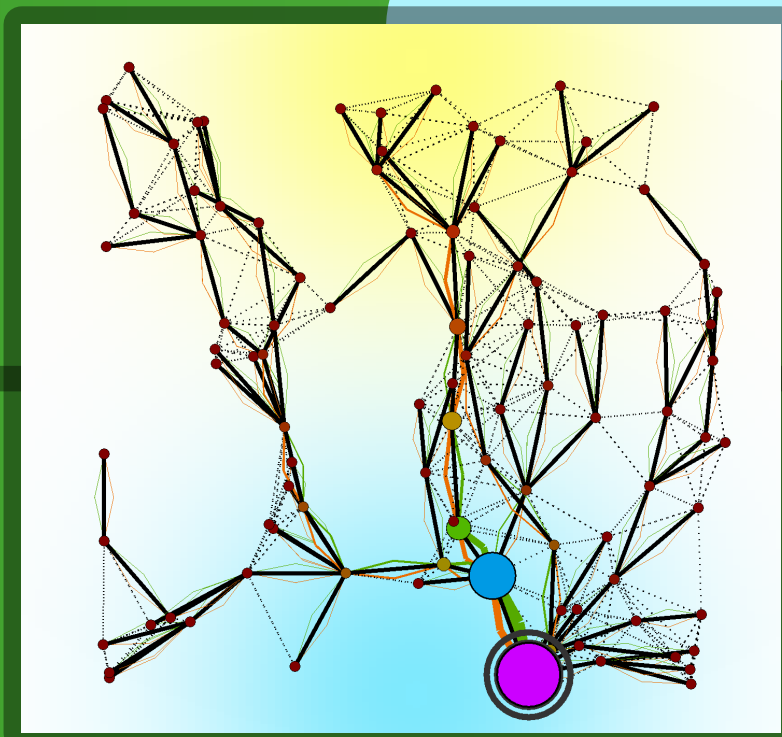
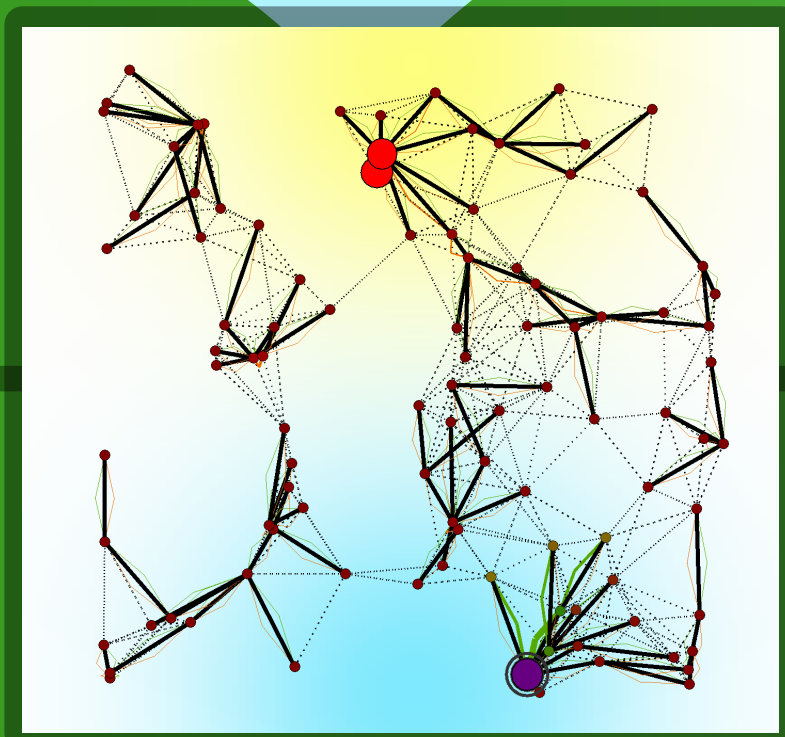
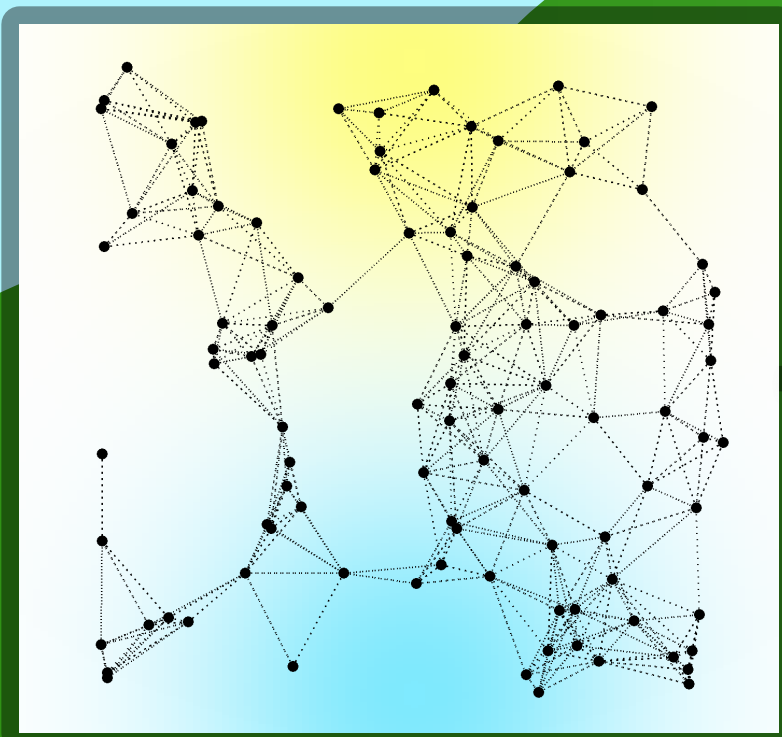




Engineered Self-Organizing Structures Through Aggregate Computing

Angela Cortecchia, Danilo Pianini, Giovanni Ciatto & Roberto Casadei -- University of Bologna



Try it yourself!

Vascular Morphogenesis Controller¹

Plant-inspired growth of artificial structures, based on the dynamics of resource distribution in tree-shaped structures (more successful leaves get more resources).



- Requires a tree structure
- Implicitly synchronous



FieldVMC

¹ Zahadat et AL. "Vascular Morphogenesis Controller: A Distributed Controller for Growing Artificial Structures."

Approach: FieldVMC

- ✓ Based on Aggregate Computing²:
 - Graphs generalization.
 - Asynchronous generalization.

² Viroli et AL. "From Distributed Coordination to Field Calculus and Aggregate Computing."

Expected benefits

- + Support to arbitrary topologies, such as having multiple roots.
- + Decentralized and asynchronous.
- + New novel phenomena like self-construction, -healing, -segmentation, -integration, -optimization

*Self-Repair (cutting):

stabilization of two new structures from a single damaged one

Self-Construction (growth from seed):
building a structure from a single node.

Self-Integration (grafting):
merging of two separate structures.

Self-Optimization (abscission and regrowth):
optimized merging of multiple structures.

Self-Segmentation (budding):
multiple independent structures from a single graph.

Experimental validation

+ Introduced with FieldVMC

